Smart Cart Industrial Development Roadmap

Author: Arpit Tiwari Target Platform:

- Jetson Orin Nano SoM
- STM32F446 + FreeRTOS
- Qt6 C++ + QML
- Eclipse Paho MQTT C++
- Embedded Linux
- Cloud Dashboard (Already Deployed)

PRODUCT VISION

The Smart Cart is an AI-powered retail cart capable of:

- One-time provisioning
- Customer onboarding
- Live shopping
- Weight-based item verification
- Cloud telemetry
- Digital checkout
- Computer vision item identification
- Item tracking
- Theft detection
- Fleet management

Architecture:

STM32F446 ↕ UART Jetson Orin Nano ↕ MQTT Cloud Dashboard

PHASE 1

ARCHITECTURE & SYSTEM DESIGN

Objective:

Freeze system architecture before implementation.

Deliverables:

- Repository Architecture

- Software Architecture
- Firmware Architecture
- UART Protocol Specification
- MQTT Strategy
- State Machine Design
- Hardware Block Diagram
- Risk Assessment

Status: COMPLETE

PHASE 2

FOUNDATION REPOSITORY

Objective:

Create compile-ready repository skeleton.

Generate:

edge-app/ firmware/ protocols/ tests/ hardware/ deployment/

Qt:

- main.cpp
- AppContext
- Controllers
- Service Interfaces
- QML Screens

STM32:

- PlatformIO
- STM32Cube
- FreeRTOS Task Skeletons

Protocols:

- Packet Structures
- CRC Interfaces

Tests:

- GoogleTest Setup

Success Criteria:

All projects compile.

PHASE 3

CORE RUNTIME FOUNDATION

Objective:

Build platform backbone.

Implement:

UART Protocol

- CRC16 CCITT-FALSE
- Packet Builder
- Packet Parser
- Command Definitions

State Machine

States:

BOOT CHECK_CONFIG PROVISIONING WAITING_APPROVAL CUSTOMER_ONBOARDING ACTIVE_SHOPPING CHECKOUT_PENDING PAYMENT_COMPLETE

Serial Service

- QSerialPort
- Packet Buffering
- Parser Integration
- Error Handling

Tests

- CRC Tests
- Packet Tests
- State Machine Tests

Success Criteria:

Reliable packet transport layer.

PHASE 4

MQTT FOUNDATION

Objective:

Establish cloud connectivity.

Implement:

MqttService

- Paho MQTT
- Reconnect Logic
- Publish Queue
- Subscription Manager

Device Configuration

device_config.json

Store:

- Device ID
- Mall ID
- Provisioned Flag
- Broker Settings

Success Criteria:

Device can persist identity and connect to cloud.

PHASE 5

DEVICE PROVISIONING

Objective:

Implement one-time mall onboarding.

Flow:

BOOT ↓ PROVISIONING ↓ WAITING_APPROVAL ↓ SAVE CONFIG ↓ REBOOT

Normal Operation:

BOOT ↓ LOAD CONFIG ↓ MQTT CONNECT ↓ CUSTOMER ONBOARDING

Rules:

Provisioning only appears:

- First Boot
- Factory Reset
- Corrupted Config
- Re-Provision Request

Success Criteria:

Provisioning never appears during normal operation.

PHASE 6

CUSTOMER SESSION SYSTEM

Objective:

Create customer lifecycle.

Screens:

- Customer Onboarding
- Shopping Screen

Workflow:

Customer enters:

Phone Number or Email

↓

Shopping Session Starts

Implement:

CustomerSession ShoppingSession

Success Criteria:

Customers can start and finish sessions.

PHASE 7

STM32 INTEGRATION

Objective:

Replace mock transport with real hardware.

Implement:

UART Packet Exchange

Commands:

CMD_HEARTBEAT CMD_WEIGHT CMD_BATTERY CMD_BUTTON CMD_LED CMD_BUZZER

FreeRTOS Tasks:

UartTask WeightTask BatteryTask LedTask WatchdogTask

Success Criteria:

Jetson and STM32 communicate reliably.

PHASE 8

CLOUD TELEMETRY

Objective:

Integrate deployed dashboard.

Publish:

Battery Status Cart Status Weight Events Device Health Customer Session State

Dashboard:

Must remain backward compatible with deployed backend.

Success Criteria:

Cart appears live on dashboard.

PHASE 9

CHECKOUT SYSTEM

Objective:

Implement shopping completion flow.

Workflow:

ACTIVE_SHOPPING ↓ CHECKOUT_PENDING ↓ PAYMENT ↓ PAYMENT_COMPLETE ↓ NEW CUSTOMER

Features:

- QR Code Payment
- Receipt Generation
- Session Closure

Success Criteria:

End-to-end shopping flow works.

PHASE 10

COMPUTER VISION FOUNDATION

Objective:

Enable AI-powered item identification.

Jetson Components:

Camera Service OpenCV Pipeline Inference Engine

Model:

YOLO

Capabilities:

Item Detection Bounding Boxes Confidence Scores

Success Criteria:

Cart can identify products using camera.

PHASE 11

TRACKING & THEFT DETECTION

Objective:

Fuse sensor and vision data.

Inputs:

YOLO Detection Weight Sensor Cart Database

Example:

Milk Detected

+ 500g Added = Valid Event

Milk Detected

+ No Weight Change = Suspicious Event

Weight Added

+ No Object Detected = Suspicious Event

Implement:

Object Tracking Sensor Fusion Theft Detection Engine

Success Criteria:

Cart can detect anomalous shopping behavior.

PHASE 12

PRODUCTION HARDWARE

Objective:

Design final carrier board.

Hardware:

Jetson Orin Nano SoM STM32F446 HX711 Battery Monitor Speaker RGB LED USB-C Debug Interfaces

Outputs:

Schematic PCB Layout BOM Gerber Manufacturing Files

Success Criteria:

Production-ready hardware.